THE INDEX DECIDES THE DOMAIN

When the radicand contains a variable, the radical becomes a function. Not every input works: the domain depends on whether the index is even or odd.

RADICAL FUNCTION

$$f(x) = \sqrt[n]{x}$$

split by **whether the index n is even or odd**

≥0

CASE 1

even index

Non-negative radicands only

starts at a point and rises gradually

E.G. $\sqrt{x}$: domain $x \geq 0$

$\mathbb{R}$

CASE 2

odd index

All real inputs

passes through the origin with an S-curve

E.G. $\sqrt[3]{x}$: domain all real numbers

TAKEAWAY

Radical functions are inverses of power functions: the square root undoes squaring, the cube root undoes cubing. See radical functions.